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CIVL2108 Principles of Civil Engineering Management

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Financial Feasibility Study-2

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1. Discount Cash Flow (DCF) Method

- **Internal rate of return (IRR)**

IRR is defined as the discount rate i that the net present worth of a project is zero if such i is used in the equivalence calculation.

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1. Discount Cash Flow (DCF) Method

End of Year	Cash out	Cash in	Net cash flow
0	C_0	B_0	$NCF_0 = B_0 - C_0$
1	C_1	B_1	NCF_1
2	C_2	B_2	NCF_2
…			
k	C_k	B_k	NCF_k
…			
n	C_n	B_n	NCF_n

The net present worth $\mathrm{NPW} = \sum_{k=0}^{n} \frac{B_k - C_k}{(1+i)^k} = \sum_{k=0}^{n} \frac{\mathrm{NCF}_k}{(1+i)^k}$

If $i = \mathrm{IRR}$, **(i.e.** $\mathrm{NPW} = 0$**)**

then $\sum_{k=0}^{n} \frac{\mathrm{NCF}_k}{(1+i)^k} = 0$

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1. Discount Cash Flow (DCF) Method

- Calculation of IRR by interpolation

A man has the opportunity of investing $9,000,000 in either Business A or Business B whose estimated incomes in the next 10 years are shown in the table below

Choose the business with higher rate of return (IRR)

Cash flow of business A

End of year	Cash out	Cash in
0	9,000,000	
1		500,000
2		800,000
3		1,200,000
4		1,500,000
5	200,000	2,000,000
6		2,000,000
7		4,200,000
8		5,000,000

Cash flow of business B

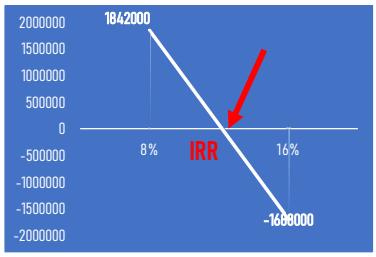
End of year	Cash out	Cash in
0	9,000,000	
1		2,500,000
2		2,300,000
3		2,000,000
4		1,500,000
5	300,000	1,500,000
6		1,000,000
7	200,000	1,000,000
8		800,000
9	200,000	800,000
10		600,000

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1. Discount Cash Flow (DCF) Method

Calculation of IRR for business A

End of year	Cash out	Cash in	Net Cash Flow (NCF)	$(DCF)^{8\%}$	$(DCF)^{16\%}$
0	9,000,000		-9,000,000	-9,000,000	-9,000,000
1		500,000	500,000	463,000	431,000
2		800,000	800,000	686,000	595,000
3		1,200,000	1,200,000	953,000	769,000
4		1,500,000	1,500,000	1,103,000	828,000
5	200,000	2,000,000	1,800,000	1,225,000	857,000
6		2,000,000	2,000,000	1,260,000	821,000
7		4,200,000	4,200,000	2,451,000	1,486,000
8		5,000,000	5,000,000	2,701,000	1,525,000
				1,842,000	-1,688000
				$(NPW)^{8\%}$	$(NPW)^{16\%}$



IRR = 12.2%

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2. Inflation

In inflation-free world

$$\text{NPW} = \frac{\text{NCF}_0}{1} + \frac{\text{NCF}_1}{(1+i)} + \frac{\text{NCF}_2}{(1+i)^2} + \ldots + \frac{\text{NCF}_n}{(1+i)^n}$$

Let f_1 = inflation expressed as a percentage in Year 1

...

f_k = inflation expressed as a percentage in Year k

...

f_n = inflation expressed as a percentage in Year n

$$\text{NPW} = \frac{\text{NCF}_0}{1} + \frac{\text{NCF}_1}{(1+i)(1+f_1)} + \frac{\text{NCF}_2}{(1+i)^2(1+f_1)(1+f_2)} + \ldots + \frac{\text{NCF}_n}{(1+i)^n(1+f_1)(1+f_2)\ldots(1+f_n)}$$

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2. Inflation

If inflation is expected to be the same in every year within the life of the project, then $f_k = f$

$$\underset{\text{(real)}}{\text{NPW}} = \frac{\text{NCF}_0}{1} + \frac{\text{NCF}_1}{(1+i)(1+f)} + \frac{\text{NCF}_2}{(1+i)^2(1+f)^2} + \ldots + \frac{\text{NCF}_n}{(1+i)^n(1+f)^n}$$

Apparent rate

$$\underset{\text{(real)}}{\text{NPW}} = \frac{\text{NCF}_0}{1} + \frac{\text{NCF}_1}{(1+i')} + \frac{\text{NCF}_2}{(1+i')^2} + \ldots + \frac{\text{NCF}_n}{(1+i')^n}$$

$$1 + i' = (1 + i)\,(1 + f)$$

Apparent rate of return $i' = (1 + i)\,(1 + f) - 1$

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2. Inflation

Example:

It is proposed to construct a self-financing road tunnel. The criterion is that the toll paid by the vehicles over the life of the project must be sufficient to cover both the initial capital and the other costs which will be incurred over the project's life. Determine a suitable charge on each vehicle passing through the tunnel using the following estimated figures based on 1996 prices.

1. Capital cost	$100,000,000
Construction cost	
Land cost	
Professional fees	
2. Annual OMR cost	$2,500,000
3. Annual administration cost	$5,000,000
4. Replacement cost of autopay ticket machine	$1,000,000 every 5 years
5. Estimated annual traffic flow	10,000 vehicles per day in the first five years; the growth rate of traffic flow is 30% every 5 years
6. Life of project	20 years
7. Required rate of return (real)	12%
8. Estimated inflation rate in the next 20 years	10% p.a. (average)

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2. Inflation

1. Initial cost	$100,000,000
2. PW of annual OMR cost	=$2,500,000 X $(uspwf)^{n=20,i=12\%}$ =18,673,500
3. PW of annual administration cost	=$5,000,000 X $(uspwf)^{n=20,i=12\%}$ =37,347,000
4. PW of replacement cost of autopay ticket machine	=$1,000,000 X $(pwf)^{n=5,i=12\%}$ + $1,000,000 X $(pwf)^{n=10,i=12\%}$ + $1,000,000 X $(pwf)^{n=15,i=12\%}$ =$1,072,080
Present worth of total costs	$157,092,580
5. Year 1- 5, no of vehicles per year = 10,000*365=3,650,000 Year 6- 10, no of vehicles per year = 10,000*365*1.3= 4,745,000 Year 10- 15, no of vehicles per year = 10,000*365*1.3*1.3= 6,168,500 Year 16- 20, no of vehicles per year = 10,000*365*1.3*1.3*1.3= 8,019,050	If y = toll paid by each vehicle Then PW of the income in year 1-5 =3,650,000 y X $(uspwf)^{n=5,i=12\%}$ Then PW of the income in year 6-10 =4,745,000y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=5,i=12\%}$ Then PW of the income in year 10- 15=6,168,500y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=10,i=12\%}$ Then PW of the income in year 16- 20 =8,019,050y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=15,i=12\%}$ Total = $35,302,558y

y = 4.45

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2. Inflation

Exam

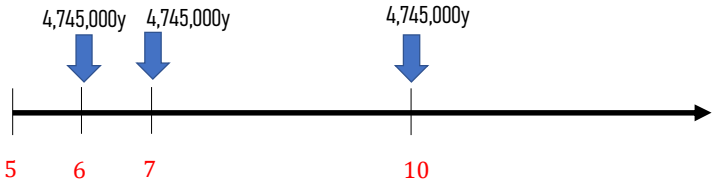
1. Initial cost	$304,000,000
2. PW of annual OMR cost	-$5,000,000 X $(uspwf)^{n=100,i=12\%}$ =18,673,500
3. PW of annual administration cost	-$2,500,000 X $(uspwf)^{n=30,i=12\%}$ =37,347,000
4. PW of replacement cost of autopay ticket machine	-$1,000,000 X $(pwf)^{n=5,i=12\%}$ + $1,000,000 X $(pwf)^{n=10,i=12\%}$ + $1,000,000 X $(pwf)^{n=15,i=12\%}$ +$1,000,000 X $(pwf)^{n=20,i=12\%}$ +$1,000,000 X $(pwf)^{n=25,i=12\%}$ =$1,072,080
Present worth of total costs	$361,092,580
5. Year 1- 5, no of vehicles per year = 10,000*365=3,650,000 Year 6- 10, no of vehicles per year = 10,000*365*1.15= 4,197,500 Year 10- 15, no of vehicles per year = 10,000*365*1.15*1.15= 4,827,125 Year 16- 20, no of vehicles per year = 10,000*365*1.15*1.15*1.15= 5,5551194	If y = toll paid by each vehicle Then PW of the income in year 1-5 =3,650,000 y X $(uspwf)^{n=5,i=12\%}$ Then PW of the income in year 6-10 =4,197,500 yX $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=5,i=12\%}$ Then PW of the income in year 10- 15=4,827,125y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=10,i=12\%}$ Then PW of the income in year 16- 20 =8,019,050y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=15,i=12\%}$ Then PW of the income in year 21- 25 =8,019,050y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=15,i=12\%}$ Then PW of the income in year 26- 30 =8,019,050y X $(uspwf)^{n=5,i=12\%}$ X $(pwf)^{n=15,i=12\%}$ Total = $35,302,558y y = 4.45

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2. Inflation

Year 6- 10



Present worth of n uniform series of payment F

$$= \frac{F}{(1+i)} + \frac{F}{(1+i)^2} + \cdots + \frac{F}{(1+i)^n}$$

$$= F\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

$$= F \cdot (\text{uspwf})^{n,i}$$

Uniform series present worth factor

At Year 5, worth of 5 uniform series of payment 4,745,000y

$$= F \cdot (\text{uspwf})^{n,i}$$

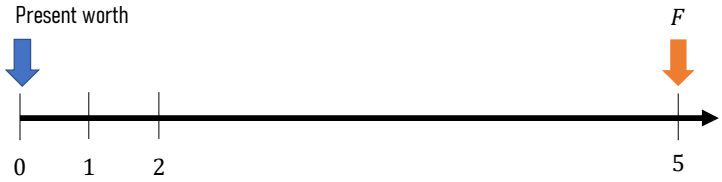
$$= 4{,}745000\text{y} \cdot (\text{uspwf})^{n=5,i=12\%}$$

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2. Inflation

Present worth of a single payment



Present worth of a single payment F at nth year

$$= \frac{F}{(1+i)^n} = F \cdot (\text{pwf})^{n,i}$$

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2. Inflation

Example:

A promoter has received two tenders for the construction of a new apartment house. The details of the tenders are:

Tender A: Contract sum $30,000,000; contract period 15 months; payment made at the end of every quarter, i.e. a total of 5 uniform payments with each payment equal to 20% of the contract sum.

Tender B: Contract sum $32,000,000; contract period 12 months; payment made at the end of every quarter, i.e. a total of 4 uniform payments with each payment equal to 25% of the contract sum.

a) Assuming that the potential rental value of the new house is $1,000,000 per month, receivable at the end of each month immediately after completion, explain by calculation why tender B should be selected by the promoter.

b) If contract B completed the contract on schedule in the first three quarters but one month late in the last quarter (so that the payments at the end of 4th quarter was 15% of the contract sum and that at the end of 13th month was 10% of the same), determine the amount that should be deducted from the final payment in order to ensure adequate compensation for the promoter.

(use the bank interest rate 12% p.a. to calculate the compensation and also use it in determining the present worth of the two tenders.)

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2. Inflation

a) **The present worths of the two tenders are:**

$$PW_A = \frac{6{,}000{,}000}{(1+0.01)^3} + \frac{6{,}000{,}000}{(1+0.01)^6} + \frac{6{,}000{,}000}{(1+0.01)^9} + \frac{6{,}000{,}000}{(1+0.01)^{12}} + \frac{6{,}000{,}000}{(1+0.01)^{15}} = 27{,}453{,}600$$

$$PW_B = \frac{8{,}000{,}000}{(1+0.01)^3} + \frac{8{,}000{,}000}{(1+0.01)^6} + \frac{8{,}000{,}000}{(1+0.01)^9} + \frac{8{,}000{,}000}{(1+0.01)^{12}} - \frac{1{,}000{,}000}{(1+0.01)^{13}} - \frac{1{,}000{,}000}{(1+0.01)^{14}} - \frac{1{,}000{,}000}{(1+0.01)^{15}} = 27{,}104{,}400$$

$PW_B < PW_A$, accept tender B

b) **Present worth of tender B for the delayed case:**

$$PW_B = \frac{8{,}000{,}000}{(1+0.01)^3} + \frac{8{,}000{,}000}{(1+0.01)^6} + \frac{8{,}000{,}000}{(1+0.01)^9} + \frac{4{,}800{,}000}{(1+0.01)^{12}} + \frac{3{,}200{,}000}{(1+0.01)^{13}} - \frac{1{,}000{,}000}{(1+0.01)^{14}} - \frac{1{,}000{,}000}{(1+0.01)^{15}} = 27{,}955{,}260$$

The amount that should be deducted from the final payment in order to ensure adequate compensation for the promoter

$$(27{,}955{,}260 - 27{,}104{,}400) \times (1 + 0.01)^{13} = \$968{,}364$$

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3. Inflation adjustment

End of Year	Cash out	Cash in	Net cash flow At actual transactions in respective years	Net cash flow at constant base year	
0	C_0	B_0	$NCF_0 = B_0 - C_0$ (based on 2010 money)	F_0 (base on 2010 money)	$NCF_0 = F_0$
1	C_1	B_1	NCF_1 (based on 2011 money)	F_1 (base on 2010 money)	$NCF_1 = F_1 \times (1+f)$
2	C_2	B_2	NCF_2 (based on 2012 money)	F_2 (base on 2010 money)	$NCF_2 = F_2 \times (1+f)^2$
…					
10	C_{10}	B_{10}	NCF_k (based on 2020 money)	F_{10} (base on 2010 money)	$NCF_{10} = F_{10} \times (1+f)^2$

$$\underset{\text{real}}{\text{NPW}} = \frac{NCF_0}{1} + \frac{NCF_1}{(1+i)(1+f)} + \frac{NCF_2}{(1+i)^2(1+f)^2} + \ldots + \frac{NCF_{10}}{(1+i)^{10}(1+f)^{10}}$$

$$= \frac{F_0}{1} + \frac{F_1 \times (1+f)}{(1+i)(1+f)} + \frac{F_2 \times (1+f)^2}{(1+i)^2(1+f)^2} + \ldots + \frac{F_{10} \times (1+f)^n}{(1+i)^{10}(1+f)^{10}}$$

$$= \sum_{k=0}^{10} \frac{F_k}{(1+i)^k}$$

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3. Inflation adjustment

The evaluation of real NPW, using NCFs of constant base year prices, has no need to consider the effects of inflation.

All methods, are valid even in a world with inflation, even if the inflation-free assumption were not made.

Most of the project appraisals are performed with their cash flows estimated at constant base year prices and so their results are real.

For financial or economic feasibility studies of project at the planning stage. One can only estimate future cash flows based on constant year prices.

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4. Present worth

- Actual transaction
- Interest rates quoted by banks are apparent rates, if the inflation rate is higher than the interest rate, it would result in a net loss for depositing money in a bank

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		20,000	80,000	10,000	30,000
2		20,000	60,000	8,000	28,000
3		20,000	40,000	6,000	26,000
4		20,000	20,000	4,000	24,000
5		20,000	0	2,000	22,000
					130,000

Method 1

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		10,000	90,000	10,000	20,000
2		15,000	75,000	9,000	24,000
3		20,000	55,000	7,500	27,000
4		25,000	30,000	5,500	30,000
5		30,000	0	3,000	33,000
					135,000

Method 2

Apparent rate $\underset{\text{(real)}}{\text{NPW}} = \frac{\text{NCF}_0}{1} + \frac{\text{NCF}_1}{(1+i')} + \frac{\text{NCF}_2}{(1+i')^2} + \ldots + \frac{\text{NCF}_n}{(1+i')^n}$

Method 1: Present worth $= \frac{30,000}{1+0.1} + \frac{28,000}{(1+0.1)^2} + \frac{26,000}{(1+0.1)^3} + \frac{24,000}{(1+0.1)^4} + \frac{22,000}{(1+0.1)^5} = \$100,000$

Method 2: Present worth $= \frac{20,000}{1+0.1} + \frac{24,000}{(1+0.1)^2} + \frac{27,000}{(1+0.1)^3} + \frac{30,000}{(1+0.1)^4} + \frac{33,000}{(1+0.1)^5} = \$100,000$

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5. Comparison of multiple alternatives ---Example

- A flood storage reservoir is to be built for a flood control scheme. Several reservoir capacities are being considered and their benefit/cost data are shown below:

	Reservoir capacity (m3)	Initial capital $ X 10^6	Annual OMR cost $ X 10^6	Annual benefit $ X 10^6
Alternative 1	50 X 10^6	50	0.4	5.6
Alternative 2	100 X 10^6	70	1.0	8.0
Alternative 3	150 X 10^6	110	1.5	10.5
Alternative 4	200 X 10^6	140	1.7	15.2
Alternative 5	250 X 10^6	180	2.0	19.0

Assuming that the project life is 30 years and that the discount rate is 6%, determine

1) The alternative with the highest net annual benefit
2) The alternative with the highest net present worth
3) The alternative with the highest B/C ratio,
4) The alternative with the highest IRR

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5. Comparison of multiple alternatives ---Example

1) The alternative with the highest net annual benefit

Alt	Equivalent annual cost of initial capital	Annual OMR cost	Total annual cost	Annual benefit	Net annual benefit
1	$50x(uscrf)^{n=30,i=60\%}$ = 3.63	0.4	4.03	5.6	+1.57
2	$70x(uscrf)^{n=30,i=60\%}$ = 5.09	1.0	6.09	8.0	+1.91
3	$110x(uscrf)^{n=30,i=60\%}$ = 7.99	1.5	9.49	10.5	+1.01
4	$140x(uscrf)^{n=30,i=60\%}$ = 10.17	1.7	11.87	15.2	+3.33
5	$180x(uscrf)^{n=30,i=60\%}$ = 13.08	2.0	15.08	19.0	+3.92

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5. Comparison of multiple alternatives ---Example

2) The alternative with the highest net present worth

Alt	initial capital	Present worth of OMR cost	Present worth of total cost	Present worth of benefits	Net present worth
1	50	0.4 x(uspwf)$^{n=30,i=60\%}$ = 5.51	55.51	5.6 x(uspwf)$^{n=30,i=60\%}$ = 77.08	+21.57
2	70	1.0 x(uspwf)$^{n=30,i=60\%}$ = 13.77	83.77	8.0 x(uspwf)$^{n=30,i=60\%}$ = 110.12	+26.35
3	110	1.5 x(uspwf)$^{n=30,i=60\%}$ = 20.65	130.65	10.5 x(uspwf)$^{n=30,i=60\%}$ = 144.53	+13.88
4	140	1.7x(uspwf)$^{n=30,i=60\%}$ = 23.40	163.40	15.2 x(uspwf)$^{n=30,i=60\%}$ = 209.23	+45.83
5	180	2.0x(uspwf)$^{n=30,i=60\%}$ = 27.53	207.53	19.0 x(uspwf)$^{n=30,i=60\%}$ = 261.53	+54.00

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5. Comparison of multiple alternatives ---Example

3) The alternative with the highest B/C ratio

Alt	Equivalent annual cost of initial capital	Annual OMR cost	Total annual cost	Annual benefit	Net annual benefit	B/C
1	50x(uscrf)$^{n=30,i=60\%}$ = 3.63	0.4	4.03	5.6	+1.57	5.6/4.03 = 1.39
2	70x(uscrf)$^{n=30,i=60\%}$ = 5.09	1.0	6.09	8.0	+1.91	1.31
3	110x(uscrf)$^{n=30,i=60\%}$ = 7.99	1.5	9.49	10.5	+1.01	1.11
4	140x(uscrf)$^{n=30,i=60\%}$ = 10.17	1.7	11.87	15.2	+3.33	1.28
5	180x(uscrf)$^{n=30,i=60\%}$ = 13.08	2.0	15.08	19.0	+3.92	1.26

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5. Comparison of multiple alternatives ---Example

Alt	initial capital	Present worth of OMR cost	Present worth of total cost	Present worth of benefits	Net present worth	B/C
1	50	0.4 x(uspwf)$^{n=30,i=60\%}$ = 5.51	55.51	5.6 x(uspwf)$^{n=30,i=60\%}$ = 77.08	+21.57	77.08/55.51=1.39
2	70	1.0 x(uspwf)$^{n=30,i=60\%}$ = 13.77	83.77	8.0 x(uspwf)$^{n=30,i=60\%}$ = 110.12	+26.35	1.31
3	110	1.5 x(uspwf)$^{n=30,i=60\%}$ = 20.65	130.65	10.5 x(uspwf)$^{n=30,i=60\%}$ = 144.53	+13.88	1.11
4	140	1.7x(uspwf)$^{n=30,i=60\%}$ = 23.40	163.40	15.2 x(uspwf)$^{n=30,i=60\%}$ = 209.23	+45.83	1.28
5	180	2.0x(uspwf)$^{n=30,i=60\%}$ = 27.53	207.53	19.0 x(uspwf)$^{n=30,i=60\%}$ = 261.53	+54.00	1.26

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5. Comparison of multiple alternatives ---Example

4) The alternative with the highest IRR

Alt	IRR
1	9.8%
2	9.3%
3	7.2%
4	8.9%
5	8.7%

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6. The incremental analysis

Neither the highest direct B/C ratio or direct IRR always corresponds to the highest NPW of multiple alternatives.

Incremental B/C ratio, defined as $\frac{\Delta B}{\Delta C}$

- Start with the project with the lowest cost and proceed in an increasing order of project cost.
- Y is better than X If $\frac{\text{Benefit of Y}-\text{Benefit of X}}{\text{Cost of Y}-\text{Cost of X}} > 1$, and vice-versa
- Repeat step 2 until all alternatives have been compared.

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6. The incremental analysis

Alt	Equivalent annual cost of initial capital	Annual OMR cost	Total annual cost	Annual benefit
1	$50x(uscrf)^{n=30,i=60\%}$ = 3.63	0.4	4.03	5.6
2	$70x(uscrf)^{n=30,i=60\%}$ = 5.09	1.0	6.09	8.0
3	$110x(uscrf)^{n=30,i=60\%}$ = 7.99	1.5	9.49	10.5
4	$140x(uscrf)^{n=30,i=60\%}$ = 10.17	1.7	11.87	15.2
5	$180x(uscrf)^{n=30,i=60\%}$ = 13.08	2.0	15.08	19.0

Alternative 1 VS Alternative 2

The incremental B/C ratio, defined as $\frac{\Delta B}{\Delta C} = \frac{8.0-5.6}{6.09-4.03} = 1.17 > 1$

Alternative 2 is better than alternative 1.

Alternative 2 VS Alternative 3

$\frac{\Delta B}{\Delta C} = \frac{10.5-8.0}{9.49-6.09} = 0.74 < 1$

Alternative 2 VS Alternative 4

$\frac{\Delta B}{\Delta C} = \frac{15.2-8.0}{11.87-6.09} = 1.25 < 1$

Alternative 4 VS Alternative 5

$\frac{\Delta B}{\Delta C} = \frac{19.0-15.2}{15.08-11.87} = 1.18 < 1$